Shopping Mall Customer Data Segmentation Analysis

Objective:
The objective of this project is to analyze customer demographics, annual income, and spending behavior to identify different customer segments. Based on the analysis, learners are expected to provide actionable business insights and marketing strategies that can help the company improve customer targeting, retention, and overall sales performance.

Part 1: Data Cleaning & Understanding

1. Import the dataset.
2. Check for missing/null values.
3. Check data types and clean the dataset.

Part 2: Customer Analysis

Answer the following:

1. What is the average age of customers?
2. What is the average annual income?
3. Which gender spends more on average?
4. Which age group has the highest spending score?
5. Is there any relationship between annual income and spending score?

Part 3: Customer Segmentation

Segment customers into groups such as:

- High Income – High Spending
- High Income – Low Spending
- Low Income – High Spending
- Low Income – Low Spending

(Advanced learners can use clustering techniques like K-Means.)

Part 4: Dashboard / Visualization

Create an interactive dashboard including:

- Gender-wise customer distribution
- Age-wise analysis
- Income vs Spending Scatter Plot
- Segment-wise distribution

Part 5: Business Recommendations

Answer:

1. Which segment should the company target most?
2. What offers can attract low-spending customers?
3. How can high-income customers be retained?

Deliverables

Dashboard / PPT / Report
Insights document
SQL / Excel / Power BI / Tableau / Python files

Plagiarism will not be accepted.